MOHAN BABU UNIVERSITY

Anveshana Club

Application Flow & Navigation UX

TARGET AUDIENCE: ENGINEERING & UX TEAMS

ALIGNED WITH PRD & TRD SPECIFICATIONS

1. Document Overview & User Types

This document defines the navigational hierarchy, interaction states, and edge-case behaviors for the Anveshana Club web platform. The platform supports two strict user typologies, with no intermediate roles or student accounts:

- **Public Visitor (Student):** Has read-only access to published content. Interactions are limited to form submissions (feedback) and external navigation (Google Forms).
- **Administrator (Core Team):** Authenticated via a hidden route. Possesses full CRUD (Create, Read, Update, Delete) permissions across all content entities.

2. Route Inventory: Public Access

Consolidation Note: The requested `/events/:slug/gallery` has been consolidated into `/events/:slug` as a dedicated on-page tab/section. This reduces unnecessary clicks and prevents orphaned gallery pages.

Route	Page / Purpose	Entry Points & Major Actions
`/`	Home	**Entry:** Direct URL. **Actions:** View mission, click featured upcoming/past events.
`/events`	Events Listing	**Entry:** Main Nav. **Actions:** Filter by Status (Upcoming/Completed), Month, Year. Click event card to view details.
`/events/:slug`	Event Details	**Entry:** Event Cards. **Actions:** Read metadata, click 'Register' (external link), view embedded image gallery if event is Completed.
`/team`	Core Team	**Entry:** Main Nav. **Actions:** Expand/collapse 7 team categories, click member social links.
`/feedback`	Feedback	**Entry:** Main Nav. **Actions:** Select 1-5 star rating, write text, submit form safely.
`/contact`	Contact Us	**Entry:** Footer. **Actions:** View official support emails, campus location map/details.
`/*` (Catch-all)	404 Not Found	**Entry:** Invalid URL. **Actions:** Click button to return to Home.

3. Route Inventory: Administrative Access

All administrative routes require a valid JWT session. Unauthenticated access immediately redirects to `/admin/login`. A successful login automatically routes to `/admin/dashboard`.

Route	Page / Purpose	Entry Points & Major Actions
`/admin` (Redirect)	Auth Gateway	Redirects to `/admin/login` if no session, or `/admin/dashboard` if authenticated.
`/admin/login`	Login	**Entry:** Direct URL. **Actions:** Submit credentials. Rate-limited to prevent brute force.
`/admin/dashboard`	Control Center	**Entry:** Post-login. **Actions:** View quick stats (total events, recent feedback), navigate to specific managers.
`/admin/events`	Events Manager	**Entry:** Admin Nav. **Actions:** List all events (table view), filter drafts vs. published, trigger deletion.
`/admin/events/new`	Create Event	**Entry:** "Add Event" button. **Actions:** Fill details, upload banner, set status, save draft or publish.
`/admin/events/:id/edit`	Edit Event	**Entry:** Event table row. **Actions:** Modify fields, manage gallery uploads (if Completed), update registration link.
`/admin/team`	Team Manager	**Entry:** Admin Nav. **Actions:** Add/edit members within strict categories, reorder via drag-and-drop.
`/admin/feedback`	Feedback Inbox	**Entry:** Admin Nav. **Actions:** View paginated list of student feedback, sort by rating/date. Read-only.

4. Core User Journeys (Public)

Journey 1: Discover the Club

Path: Home `/` → Scroll Sections → Event Details `/events/:slug`.

Flow: Visitor lands on the cinematic hero section. They scroll past the mission statement to the "Upcoming Events" carousel. They click a featured event card. The application transitions smoothly to the detailed event view, presenting the full description and venue map.

Journey 2: Register for an Event

Path: Events Listing `/events` → Select Event → Click CTA → External Link.

Flow & States: Visitor filters events to "Upcoming". Opens an event. The UI behavior relies entirely on the event's `status` and `registrationLink` fields:

- *Status: Upcoming (No URL)* - CTA is grayed out, reads "Registrations Opening Soon".
- *Status: Open (Valid URL)* - CTA is Golden-Orange, reads "Register Now". Clicking it opens the Google Form in a new browser tab (`_blank`), preserving the Anveshana site in the original tab.
- *Status: Closed/Completed* - CTA is disabled, reads "Registrations Closed".

Journey 3: Explore Past Events & Galleries

Path: Events Listing `/events` → Filter: Completed → Open Event → View Gallery.

Flow: Visitor applies the "Completed" filter. Opens a past Hackathon. Below the event description, a masonry grid of uploaded photographs is displayed. Clicking a thumbnail opens a full-screen lightbox overlay allowing left/right navigation through the gallery.

Journey 4: Explore the Core Team

Path: Global Nav → Team `/team` → Expand Category → View Profiles.

Flow: Visitor opens the Team directory. Categories (e.g., Design Team, Technical Team) are presented as accordions or distinct scrolling sections. Expanding a section reveals member cards. If a photo fails to load or is missing, a unified Anveshana monogram placeholder is rendered locally.

5. Core User Journeys (Administrative)

Journey 5: Administrator Login & Session Management

Path: Manual entry of `/admin` → Redirect `/admin/login` → Authenticate → Dashboard.

Flow: Admin inputs credentials. On success, an HttpOnly cookie is set, and React Router pushes to `/admin/dashboard`. If credentials are invalid, an inline red text error appears. If the session expires during active use, the next API call returns 401 Unauthorized, triggering an automatic global redirect back to `/admin/login?expired=true` with a toast notification.

Journey 6: Event Creation Workflow

Path: Dashboard → `/admin/events/new` → Form Input → Preview → Publish.

Flow: Admin fills standard fields (Title, Date). They drag-and-drop a banner image; the UI shows a loading spinner during Cloudinary upload and replaces it with a thumbnail upon success. Admin pastes the Google Form URL and sets status to "Open". Before saving, they click "Preview" to see a modal reflecting the public view. They click "Publish", returning them to the event table with a success toast.

Journey 7: Core Team Management

Path: Dashboard → `/admin/team` → Add Member → Assign Category → Save.

Flow: Admin selects "Add Member". They must select a predefined category (e.g., PR Team) from a locked dropdown—free text is forbidden to maintain schema integrity. They input URLs for LinkedIn/GitHub. If URLs are malformed, Zod validation blocks submission. On save, the public `/team` page is instantly updated.

Journey 8: Feedback Moderation

Path: Dashboard → `/admin/feedback` → Read Submissions.

Flow: Admin views a data table of feedback. To prevent accidental data loss, feedback cannot be edited. A "Delete" action is available behind a confirmation modal ("Are you sure? This cannot be undone.") for removing spam.

6. Architecture & Logic Diagrams

The following diagrams are provided in standard Mermaid syntax for parsing by development teams and AI coding agents. They define strict state transitions and routing architectures.

Diagram 1 & 2: Sitemap & Visitor Navigation

```
%% Diagram 1: Overall Sitemap & Route Hierarchy
graph TD
    Root[/ anveshana.mbu.edu /] --> Pub[Public Routes]
    Root --> Adm[Admin Routes /admin]

    Pub --> H[Home /]
    Pub --> E[Events /events]
    Pub --> T[Team /team]
    Pub --> F[Feedback /feedback]
    Pub --> C[Contact /contact]

    E --> ED[Event Detail /events/:slug]

    Adm --> AL[Login /admin/login]
    Adm --> AD[Dashboard /admin/dashboard]
    AD --> AE[Manage Events]
    AD --> AT[Manage Team]
    AD --> AF[Review Feedback]

    AE --> AEC[Create /events/new]
    AE --> AEE[Edit /events/:id/edit]

%% Diagram 2: Public Visitor Navigation
stateDiagram-v2
    [*] --> Home
    Home --> EventsListing : Click "Events"
    Home --> TeamDirectory : Click "Team"
```

Diagram 3 & 4: Registration & Auth Flows

```
%% Diagram 3: Event Registration Journey
flowchart TD
    A[Visitor on Event Details] --> B{Check Event Status}
    B -- Upcoming --> C[Button: Registrations Opening Soon]
    C --> |Disabled| A
    B -- Closed/Completed --> D[Button: Registrations Closed]
    D --> |Disabled| A
    B -- Open --> E{Check Registration URL}
    E -- Null/Empty --> F[Button: Link Unavailable]
    F --> |Disabled| A
    E -- Valid URL --> G[Button: Register Now]
    G --> |Click| H[Open New Tab: Google Forms]

%% Diagram 4: Administrator Authentication Flow
sequenceDiagram
    participant Admin
    participant Frontend
    participant API
    participant DB

    Admin->>Frontend: Access /admin/dashboard
    Frontend->>API: GET /api/v1/auth/me (Check Session)
    API-->>Frontend: 401 Unauthorized
    Frontend-->>Admin: Redirect to /admin/login

    Admin->>Frontend: Enter Email & Password
    Frontend->>API: POST /api/v1/auth/login
    API->>DB: Fetch Hash & Compare
```

Diagram 5 & 6: Workflow & Error Handling

```
%% Diagram 5: Event Creation and Publishing Workflow
stateDiagram-v2
    [*] --> FormInput : Click "Add Event"
    FormInput --> ImageUpload : Drop Banner Image
    ImageUpload --> Uploading : Contact Cloudinary
    Uploading --> ImageUploadError : Failed
    Uploading --> FormInput : Success (Thumbnail Appears)

    FormInput --> Validation : Click Save
    Validation --> FormInput : Zod Error (Missing Fields)
    Validation --> DBWrite : Pass Validation

    DBWrite --> SavedDraft : Status = Draft
    DBWrite --> Published : Status = Open/Upcoming

    SavedDraft --> [*]
    Published --> PublicSite : Instantly Visible
    PublicSite --> [*]

%% Diagram 6: Error Handling & Redirects
flowchart TD
    Req[Incoming Request] --> RouteCheck{Route Exists?}
    RouteCheck -- No --> FOF[Render 404 Component]
    RouteCheck -- Yes --> AuthCheck{Is Admin Route?}

    AuthCheck -- No --> Fetch[Fetch Public Data]
    Fetch -- DB Down --> Err[Render 500 Skeleton/Toast]
    Fetch -- Success --> Render[Render View]
```

7. Interaction Specifications

7.1 Public Interface Interactions

Action Trigger	Preconditions	Result & State Changes	Edge / Error Handling
Load Events (Page Mount)	None.	Displays Skeleton Loader. On success, maps array to Event Cards.	If DB fails, show generic "Unable to load events" card. If empty array, show "No events scheduled".
Click 'Register' (Event Detail)	Event status == 'Open' AND valid URL exists.	Opens `target="_blank"` linking to Google Forms. Original tab remains intact.	If status changes to Closed while user is on page, button is visually disabled, click blocked.
Submit Feedback	Rating > 0 selected. Text < 500 chars.	Button swaps to spinner. POST to API. On success, form swaps to "Thank You" message component.	Rate limit hit: Show "Please wait before submitting again." Network fail: Toast error message.
Expand Team (Accordion click)	Team categories fetched.	Smooth CSS height transition. Arrow icon rotates 180deg. Renders member cards.	If category array is empty, render text "Team members updating soon."
Click Back (Browser native)	User is deep inside an event.	Restores previous route. React Query restores cached scroll position for listing.	N/A (Standard DOM behavior).

7.2 Administrative & Form Interactions

Action Trigger	Preconditions	Result & State Changes	Edge / Error Handling
Upload Banner (Drag & Drop)	File is jpeg/png/webp, < 2MB.	XHR to backend proxy → Cloudinary. UI shows progress bar. Replaces dropzone with image preview.	Size limit exceeded: Red text "File too large". Format error: "Invalid format".
Navigate Away (Dirty Form)	React Hook Form `isDirty == true`.	Intercept routing. Display modal: "You have unsaved changes. Leave page?"	User clicks "Cancel": Stay on form. Click "Leave": Reset form state and route.
Click Publish (Event Form)	All mandatory Zod fields valid.	Button state: `isSubmitting`. POST to API. Redirect to `/admin/events`, show success toast.	API returns 401 (Session Expired): Redirect to login, form data is lost (expected security behavior).
Delete Event (Table Action)	User clicks trash icon.	Opens confirmation modal. If confirmed, DELETE call sent. Row fades out of table.	Backend fails to delete Cloudinary asset: DB record still deleted, logs warning on server. Toast says "Deleted".

8. Comprehensive Edge Cases

> **1. Missing Media Assets**
>
> If an event banner or team member photograph URL returns a 404 (or is omitted entirely), the frontend component must catch the `onError` native image event and swap the `src` to a local, bundled fallback image (e.g., an elegant monogram of the Anveshana logo) to prevent broken UI layouts.

> **2. The "Ghost Event" Scenario**
>
> An event is marked "Upcoming" but the date has physically passed in real time. The frontend should *not* attempt to dynamically override backend truth based on the user's clock, as timezones vary. The UI faithfully renders what the DB says. The Admin must manually update the status to "Completed".

> **3. Unauthorized API Penetration**
>
> If a savvy user intercepts API calls and attempts a POST to `/api/v1/events` without a valid JWT cookie, the Express backend immediately terminates the request at the middleware level, returning `401 Unauthorized` without executing any DB queries or Cloudinary upload streams.